03. Exploratory Data Analysis

Understand target balance, distributions and feature relationships.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

EDA actually performed

- Inspected head(), info(), isnull().sum() and describe().
- Plotted target class proportions using a pie chart.
- Plotted Education_Level category counts.
- Viewed applicant and co-applicant income histograms.
- Compared applicant income, credit score, DTI ratio and savings with boxplots split by approval class.
- Plotted credit-score and applicant-income histograms by target class.
- After encoding, plotted a numeric correlation heatmap and sorted correlations with Loan_Approved.

Observed class balance

After most-frequent imputation of the target, the notebook's saved outputs show 652 No and 348 Yes (65.2% / 34.8%). This is not the verified class balance among originally labelled rows because 50 unknown target values were filled with the mode.

Correlation output

Feature	Correlation with encoded target (not causal)
Credit_Score	+0.451
Applicant_Income	+0.120
Employer_Category_MNC	+0.069
Collateral_Value	+0.022
Coapplicant_Income	+0.004
Existing_Loans	–0.035
DTI_Ratio	–0.0 approx; value not clearly surfaced
Education_Level	inspect notebook output for exact value

Interpretation and caution

Credit_Score is the strongest simple linear association in the notebook's correlation table. This does not prove that credit score causes approval or that it will generalize. Correlation is sensitive to the encoded class mapping and does not capture non-linear effects or interactions.

EDA improvements

- Repeat all target-balance and feature-vs-target analysis using only rows with known target labels.
- Inspect missingness co-occurrence and whether the same 50 rows are missing across all fields.

- Add missingness heatmaps, duplicate checks, distribution summaries and domain validation.
- Use grouped approval rates and confidence intervals, not only histograms.
- Review potential proxy variables and disparate impact by legally appropriate groups; do not infer fairness from aggregate accuracy.